Lecture 3

Tuesday AM, 7/14

Topics/outline

Transverse oscillations continued

- Matrix representation
- Thin magnet limit
- Stability
- Courant-Snyde parametrization
- If time: closed form periodic solution

Describing transverse motion

Lecture 2 recap:

- Transverse plane: particles undergo betatron oscillations around an ideal/design path
  - ↳ ideal closed orbit for circular acc.
- Equations of the form (Hill eq.)

$$\underbrace{x''}_{\frac{d^2x}{ds^2}} + K_x(s)\,x = 0, \quad z'' + K_z(s)\,z = 0$$

- K focusing functions of $s \rightarrow$ path along ideal orbit (curvilinear coordinates)

Solutions $x(s)$, $z(s)$?

Generalize to an arbitrary coordinate

$$u(s) \Rightarrow u'' + K u = 0$$

- Will find solutions of the form

$$u(s) = A\underbrace{\sqrt{\beta(s)}}_{\text{amplitude}} \cos\big(\underbrace{\psi(s)}_{\text{phase}} - \psi_0\big)$$

☆ Throughout this lecture $\beta = \beta(s)$ amplitude function, NOT relativistic β

- cos term with a phase advance, but also $\beta(s)$ term ⇒ varying amplitude function
  - ↳ β vs s: key part of lattice design
- How to find solutions $u(s)$?
- Two possible approaches:
  - Piecewise, where K is constant within a given element
  - Analytical solution
- We'll start with piecewise-approach, help build transport matrix description, describe individual magnets, introduce CS parameterization
- Then: discuss analytical process
  - ↳ easier to solve with periodic boundaries, but not required, see H.W. 8.2

Piecewise approach

S.Y. Lee 2.II.1
H.W. 5.5.1, 7.2

- K is constant ⇒ harmonic oscillator
  - ↳ good approximation within a given element (magnet)
  - ↳ solutions of the form of sin, cos

$C(s)$ = cosine-like $S(s)$ = sine-like

$$u(s) = C(s)\,u_0 + S(s)\,u_0'$$
$$u'(s) = C'(s)\,u_0 + S'(s)\,u_0'$$

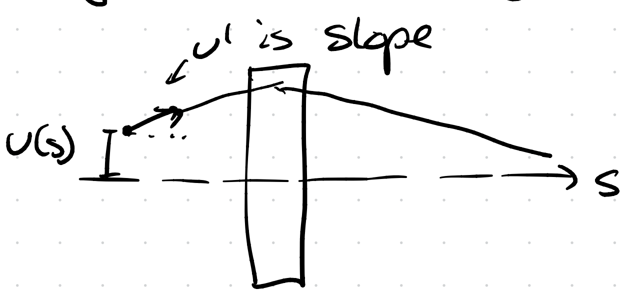
u_0, u_0' are initial coordinates

* What is u'? $\frac{du}{ds}$, slope of particle's trajectory, u and u' describe the phase space

↓ see lec 1

$x' = p_x/p$

e.g. when focusing



· Satisfy initial conditions:

$$u(s=s_0) = u_0 \Rightarrow \begin{array}{l} C(s=s_0)=1 \\ S(s=s_0)=0 \end{array}$$

$$u'(s=s_0) = u_0' \Rightarrow \begin{array}{l} C'(s=s_0)=0 \\ S'(s=s_0)=u_0' \end{array}$$

Can express these equations as a matrix:

$$\begin{bmatrix} u(s) \\ u'(s) \end{bmatrix} = \underbrace{\begin{bmatrix} C(s) & S(s) \\ C'(s) & S'(s) \end{bmatrix}}_{\text{transfer matrix } M} \begin{bmatrix} u_0 \\ u_0' \end{bmatrix}$$

· Propagate u, u' through some element from $s_1 \rightarrow s_2$ (constant K within element)

$$\begin{bmatrix} u(s_2) \\ u'(s_2) \end{bmatrix} = M_{(s_2|s_1)} \begin{bmatrix} u(s_1) \\ u'(s_1) \end{bmatrix}$$

- Solutions for our elements so far?
- Start simple: drift space, no fields
- Recall from Lec 2, for linear magnets + uncoupled transverse coordinates, field had the form

$$B_x(s) = \frac{\partial B_z}{\partial x} z \qquad B_z(s) = -B_0 + \frac{\partial B_z}{\partial x} x$$

Gave us K of the form

$$K_x(s) = \frac{1}{\rho^2} + K_1(s), \qquad K_z(s) = -K_1(s)$$

- No fields $\Rightarrow$ $K=0$ special case

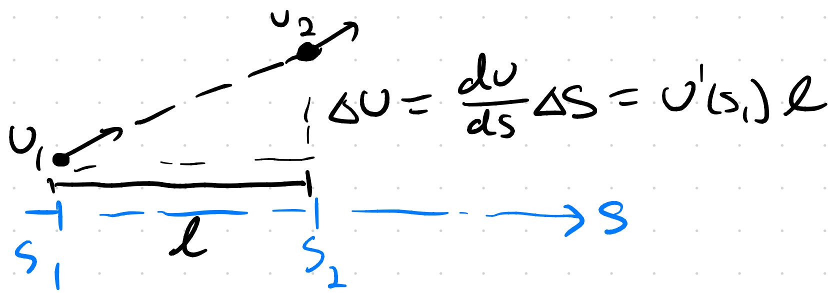
For a drift length $l = s_2 - s_1$

$$\boxed{M_{(s_2|s_1)} = \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix}}$$

* check this makes sense!

$$\begin{bmatrix} u_2 \\ u_2' \end{bmatrix} = \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_1' \end{bmatrix} \Rightarrow \boxed{\begin{aligned} u_2 &= u_1 + l u_1' \\ u_2' &= u_1' \end{aligned}}$$

- Slope of trajectory (divergence) is unchanged
- Position changes based on divergence



Solutions for $K > 0$ (focusing)

$$C(s) = \cos(\sqrt{K}\, s) \qquad S(s) = \frac{1}{\sqrt{K}}\sin(\sqrt{K}\, s)$$

Find

$$C'(s) = -\sqrt{K}\sin(\sqrt{K}s)$$
$$S'(s) = \cos(\sqrt{K}s)$$

Again describe path $l = s - s_0$ or $s_2 - s_1$

$$\boxed{M = \begin{bmatrix} \cos(\sqrt{K}\, l) & \frac{1}{\sqrt{K}}\sin(\sqrt{K}\, l) \\ -\sqrt{K}\sin(\sqrt{K}\, l) & \cos(\sqrt{K}\, l) \end{bmatrix}}$$

Solutions for $K<0$ (defocusing)

$$C(s) = \cosh(\sqrt{|K|}\, s), \quad S(s) = \frac{1}{\sqrt{|K|}} \sinh(\sqrt{|K|}\, s)$$

$$\Rightarrow C'(s) = \sqrt{|K|} \sinh(\sqrt{|K|}\, s)$$ ⇐ note sign!

$$S'(s) = \cosh(\sqrt{|K|}\, s)$$

Through path length of ℓ

$$\boxed{M = \begin{bmatrix} \cosh(\sqrt{|K|}\,\ell) & \frac{1}{\sqrt{|K|}} \sinh(\sqrt{|K|}\,\ell) \\ \sqrt{|K|} \sinh(\sqrt{|K|}\,\ell) & \cosh(\sqrt{|K|}\,\ell) \end{bmatrix}}$$

Note sign of bottom right term

$\Rightarrow$ significance?

$$u(s) = C(s)\, u_0 + S(s)\, u_0'$$

$$u'(s) = \underbrace{C'(s)}_{\text{decreasing vs increasing slope}} u_0 + S'(s)\, u_0'$$

* see also H.W. 7.4.1

relationship to focal length generally

Will be clearer in thin lens case

Before we look at that, one more special case:

Sector dipole: $K = \frac{1}{\rho^2}$ ($K>0$ sols.)

$$C(s) = \cos(\sqrt{K}s) = \cos(s/\rho)$$

$$S(s) = \frac{1}{\sqrt{K}}\sin(\sqrt{K}s) = \rho\sin(s/\rho)$$

$$l = s - s_0 = \theta/\rho \text{ (as before)}$$

$$\boxed{M = \begin{bmatrix} \cos\theta & \rho\sin\theta \\ -\frac{1}{\rho}\sin\theta & \cos\theta \end{bmatrix}}$$

bend focusing mentioned yesterday

Looking at both coordinates (x, z in S.Y. Lee; x, y in H.W)

$$\begin{bmatrix} x \\ x' \\ z \\ z' \end{bmatrix}_{s=s_2} = \begin{bmatrix} M_x(s_2|s_1) & 0 \\ 0 & M_z(s_2|s_1) \end{bmatrix} \begin{bmatrix} x \\ x' \\ z \\ z' \end{bmatrix}_{s=s_1}$$

off-diagonal terms are 0
transverse dimensions uncoupled

- When working with x, z remember two forms of K for our linear magnets

No bending in vertical plane

$$K_x(s) = \frac{1}{\rho^2} + K_1(s), \quad K_z(s) = -K_1(s)$$

quadrupole: focusing in one defocusing in other

Thin magnets / magnetic optics

H.W. 7.2.3

- sector dipole: $\theta = l/\rho$, $l \ll \rho$ take small angle limit $\cos\theta = 1$, $\sin\theta = \theta$

$$M = \begin{bmatrix} \cos\theta & \rho\sin\theta \\ -\frac{1}{\rho}\sin\theta & \cos\theta \end{bmatrix} \Rightarrow M = \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix}$$

- $l \ll \rho$ sector dipole looks like a drift (remember s is our curvilinear coordinate along orbit!)

- Thin quadrupole $l \to 0$ (focusing)

$$M = \begin{bmatrix} \cos(\sqrt{K}\, l) & \frac{1}{\sqrt{K}} \sin(\sqrt{K}\, l) \\ -\sqrt{K} \sin(\sqrt{K}\, l) & \cos(\sqrt{K}\, l) \end{bmatrix}$$

- Define "focal length" $\frac{1}{f} = K l$

$\left(K_{quad} = K_1 \text{ only}, \; K_1 = \pm \frac{B_1(s)}{B\rho} \right)$

- Take limits of each term $l \to 0$
  ↳ Taylor expansion!!

- Get thin lens form: $M = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix}$

Focusing:

$$M_F = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix}$$

Defocusing:

$$M_D = \begin{bmatrix} 1 & 0 \\ \frac{1}{f} & 1 \end{bmatrix}$$

note sign

- Think of "thin lens" the same as in geometric optics $f \gg l$

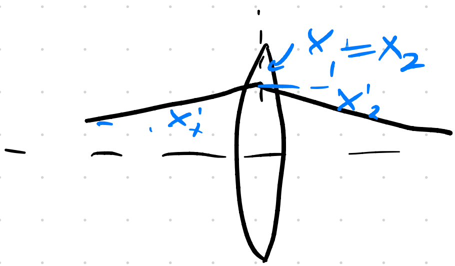
Example: thin lens focusing

$$\begin{bmatrix} x \\ x' \end{bmatrix}_2 = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \begin{bmatrix} x \\ x' \end{bmatrix}_1$$

$x_2 = x_1$ ← no change in coordinate

$x_2' = -\frac{1}{f} x_1 + x_1'$

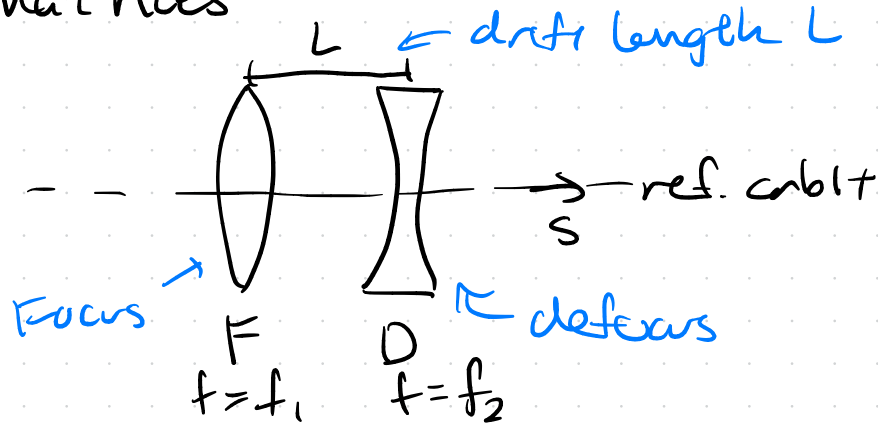
↑ change in angle



Thin lens

Know focusing in one plane is defocusing in the other, but we can achieve focusing in both planes by alternating F/D ⇒ geometric optics principle

We will show this using transfer matrices



$\Rightarrow$ Thin lens doublet

closest to $\begin{bmatrix} x \\ x' \end{bmatrix}_1$

$$M_{Doublet} = M_D \cdot M_{Drift} \cdot M_F$$

$$\begin{bmatrix} x \\ x' \end{bmatrix}_2 = \begin{bmatrix} 1 & 0 \\ \frac{1}{f_2} & 1 \end{bmatrix} \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\frac{1}{f_1} & 1 \end{bmatrix} \begin{bmatrix} x \\ x' \end{bmatrix}_1$$

* you'll need to remember how to multiply matrices

$A \cdot (B \cdot C) = (A \cdot B) \cdot C$ but $A \cdot B \neq B \cdot A$

Find $M_{Doublet}$

$$M_{Doublet} = \begin{bmatrix} 1 & 0 \\ \frac{1}{f_2} & 1 \end{bmatrix} \begin{bmatrix} 1 - \frac{L}{f_1} & L \\ -\frac{1}{f_1} & 1 \end{bmatrix}$$

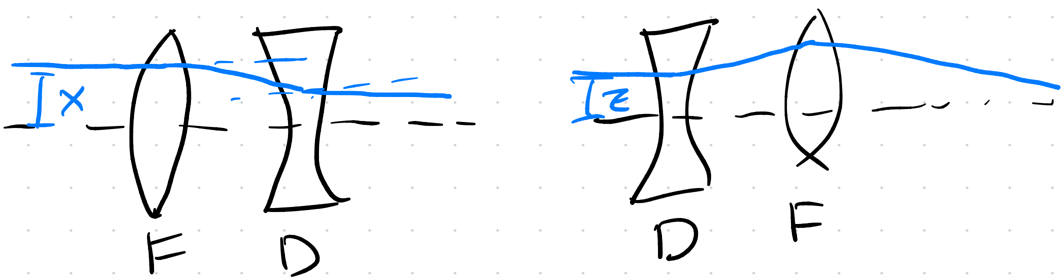
$$= \begin{bmatrix} 1 - L/f_1 & L \\ \underbrace{\frac{1}{f_2}\left(1 - \frac{L}{f_1}\right) - \frac{1}{f_1}}_{\text{focusing term}} & \frac{L}{f_2} + 1 \end{bmatrix}$$

$$f_{Doublet} = \frac{1}{f_2} - \frac{L}{f_2 f_1} - \frac{1}{f_1}$$

If equal strengths, $f_2 = f_1$

$$\boxed{f_{\text{Doublet}} = \frac{-L}{f^2}} \text{ net focusing!}$$

Easiest to visualize in paraxial case



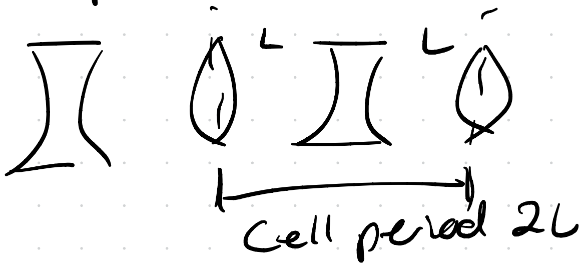
Reverse order: same doublet

"FODO" focus-defocus

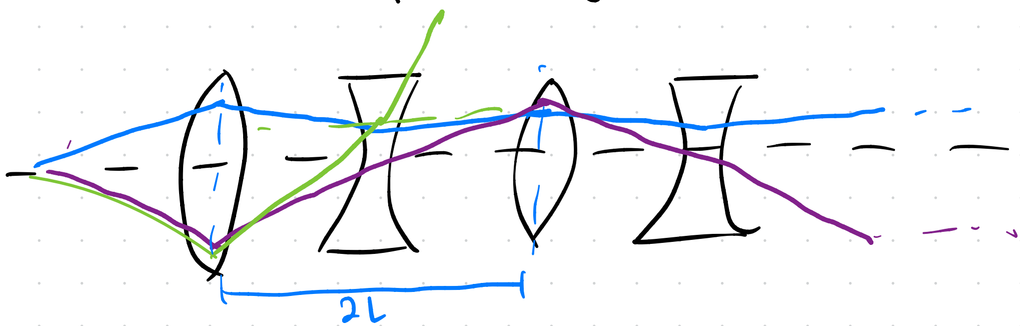
- Accelerator lattice consists of many
- Consider periodic case, like a synchrotron
  - ↳ what are stable orbits?

Stability + periodic cases

- Example of FODO lattice: define period $= 2L$ (drifts L in between)



- Can imagine many cases for betatron oscillations depending on f, L



What determines stability in periodic case? (either periods of lattice elements, turns of accelerator)

Transfer matrix for N elements

$$M = M_N M_{N-1} \cdots M_2 M_1$$

- Let's say this is transfer matrix of one turn of circular accelerator like a synchrotron

- Stability in periodic condition means that for n turns

*S.Y. Lee 2.II.1
H.W. 10.1.2

$$\begin{bmatrix} x \\ x' \end{bmatrix}_n = \underbrace{M^n}_{\text{must remain finite for large } n} \begin{bmatrix} x \\ x' \end{bmatrix}_{in}$$

- Establish some restriction on M

- Consider complex eigenvectors $\vec{v}_1, \vec{v}_2$ of matrix M, two complex eigenvalues λ_1, λ_2 that satisfy $M\vec{v} = \lambda\vec{v}$

- Write our initial condition in this basis

$$\vec{x}_{in} = A\vec{v}_1 + B\vec{v}_2 \quad \left(* \; \vec{x}_{in} = \begin{bmatrix} x \\ x' \end{bmatrix}_{in}\right)$$

complex constants

$$\Rightarrow \vec{X}_{out} = M^n \vec{X}_{in} = A\lambda_1^n \vec{V}_1 + B\lambda_2^n \vec{V}_2$$

λ_1^n and λ_2^n must remain finite for arbitrarily large n

- Find λ_1, λ_2 via

Identity matrix

$$\det(M - \lambda I) = 0$$

M of the form $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$M - \lambda I = \begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix}$$

$$\Rightarrow \det(M - \lambda I) = (a-\lambda)(d-\lambda) - bc = 0$$

Rearrange to get characteristic eq.

$$\underbrace{(ad-bc)} - a\lambda - d\lambda + \lambda^2 = 0$$

$$\det(M) = ad - bc$$

- Note we constructed our M from $C(s)$ and $S(s)$ ⇒ two linearly independent solutions to Hill eq

⇒ special property, M is unimodular (H.W. 5.5.3)

$$\det(M) = 1$$

$$M = \begin{vmatrix} C(s) & S(s) \\ C'(s) & S'(s) \end{vmatrix} = C(s)S'(s) - C'(s)S(s)$$

* $\cos^2 + \sin^2 = 1$

- Note $\det(AB) = \det(A) \times \det(B)$

square matrices, same dimensions

$$\det(M) = \det(M_N) \cdots \det(M_1) = 1$$

⇒ characteristic eq becomes

$$(1 - (a+d)\lambda + \lambda^2 = 0) \; 1/\lambda$$

$$\lambda^{-1} + \lambda = a + d = \mathrm{Tr}(M)$$

← Trace of M

- Unimodular matrix ⇒ eigenvalues are reciprocals of each other $\lambda_2 = 1/\lambda_1$

$$\lambda_1 + \lambda_2 = Tr(M) = C(s) + S'(s)$$

Solutions of the form of complex exp. satisfy this

$\lambda_1 = e^{i\phi}$, $\lambda_2 = e^{-i\phi}$ gives

$$\lambda_1 + \lambda_2 = \cos\phi + i\sin\phi + \cos\phi - i\sin\phi = 2\cos\phi$$

$$\cos\phi = \frac{1}{2}\left(C(s) + S'(s)\right)$$

$$2\cos\phi = Tr(M)$$

$\lambda_1 = e^{i\phi}$ Need ϕ to be real, real when

$\boxed{|Tr(M)| \leq 2}$ $\Rightarrow$ requirement for stability

ϕ is called the betatron phase (phase of betatron oscillation)

What restriction does this set for periodic FODO cell? $\Rightarrow$ Homework

Courant-Snyder Parametrization

S.Y. Lee II.2
H.W. 10.1.2

- Can write M in form
  ↳ still unimodular

$$M = \begin{pmatrix} \cos\phi + \alpha\sin\phi & \beta\sin\phi \\ -\gamma\sin\phi & \cos\phi - \alpha\sin\phi \end{pmatrix}$$

← betatron phase advance

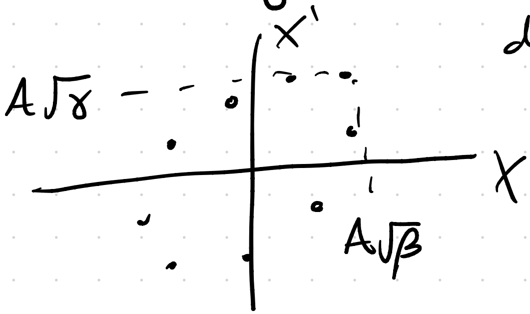
$$= \underset{\leftarrow \text{Identity matrix}}{I}\cos\phi + J\sin\phi$$

where: $J = \begin{bmatrix} \alpha & \beta \\ -\gamma & -\alpha \end{bmatrix}$ and $\gamma \equiv \dfrac{1+\alpha^2}{\beta}$

- β, α, and γ are the Courant-Snyder or Twiss parameters (or betatron functions)
  all are functions of s

- β is related to amplitude of oscillation $\Rightarrow$ amp $\propto \sqrt{\beta(s)}$
- $\alpha = -\frac{1}{2}\beta'$
- Together they describe an ellipse in transverse phase space

- As Doug mentioned in Lec. 1, describe both beam "envelope" as well as single particle



describe shape of this ellipse

- Here just introduced as convenient parametrization of M (for now)
- Can use in form $\vec{x} = M \vec{x}_{in}$, but can also describe how $\beta(s), \alpha(s), \gamma(s)$ evolve through lattice: S.Y. Lee 2.II.2, H.W. 8.1.2

For lattice with

$$M = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}$$

$$\begin{bmatrix} \beta(s) \\ \alpha(s) \\ \gamma(s) \end{bmatrix}_2 = \begin{bmatrix} m_{11}^2 & -2m_{11}m_{12} & m_{12}^2 \\ -m_{11}m_{21} & m_{11}m_{22}+m_{12}m_{21} & -m_{12}m_{22} \\ m_{21}^2 & -2m_{21}m_{22} & m_{22}^2 \end{bmatrix} \begin{bmatrix} \beta(s) \\ \alpha(s) \\ \gamma(s) \end{bmatrix}_1$$

Closed-form periodic solution

S.Y. Lee 2.II.3, H.W. 10.2.1

- Did piece-wise solution, can instead find closed-form solution in periodic case

$$u'' + K(s)u = 0$$

Require K to be periodic around one turn $K(s+L) = K(s)$

L_p is our period of interest, e.g. one turn of a circular accelerator

L_p = circumference

see S.Y. Lee Appendix A.I.5

- Floquet's Theorem:

Can write solutions of the form

$$u(s) = a\,w(s)\,e^{\pm i\psi(s)}$$

$$w(s+L) = w(s)$$

$w(s)$ amplitude

$\psi(s)$ phase

$$\psi(s+L)-\psi(s)=2\pi\mu$$ phase advance per period

- Derivations too much for this
  - ↳ Use M to find w that satisfy periodic condition
  - ↳ use M in CS form to relate to Twiss parameters:

$$\beta(s)=\frac{w^2(s)}{a}$$ ← constant

S.Y. Lee chooses choose a →
$$\beta(s)=w^2(s)$$

Weidemann $a=\frac{U_0}{\sqrt{\beta_0}}$ initial conditions

$$\psi(s+L)-\psi(s)=\Phi$$ phase advance over period

- SY Lee generalizes from here to any beam transport
- H.W. derives in non-periodic case initially with ansatz 8.2–8.3

· Result is the same form

$$u(s) = a\sqrt{\beta(s)}\cos\left(\psi(s) + \xi\right)$$

initial conditions

General transfer matrix: S.Y. Lee eq 2.42
H.W 8.74

$$M(s|s_0) = \overbrace{\begin{bmatrix} C & S \\ C' & S' \end{bmatrix}}^{\text{Wiedemann}} = \overbrace{\begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}}^{\text{SY Lee}}$$

$s_0 \rightarrow s$

$$C = \sqrt{\frac{\beta}{\beta_0}}\left(\cos\psi + \alpha_0 \sin\psi\right) = m_{11}$$

$$S = \sqrt{\beta\beta_0}\,\sin\psi = m_{12}$$

$$C' = \frac{\alpha_0 - \alpha}{\sqrt{\beta\beta_0}}\cos\psi - \frac{1+\alpha\alpha_0}{\sqrt{\beta\beta_0}}\sin\psi = m_{21}$$

$$S' = \sqrt{\frac{\beta_0}{\beta}}\left(\cos\psi - \alpha\sin\psi\right) = m_{22}$$

☆ $\psi = \psi(s_0 \rightarrow s)$ phase advance

· Phase is related to amplitude / β by

↙ or $\Delta\psi$

$$\psi(s_0 \to s) = \int_{s_0}^{s} \frac{ds}{\beta(s)}$$

Note for periodic matching
$\beta_0 = \beta$, $\alpha = \alpha_0$, reproduce
CS parametrization matrix above!

Can also write in the form

$$M(s|s_0) = B(s|s_0) \begin{bmatrix} \cos\psi & \sin\psi \\ -\sin\psi & \cos\psi \end{bmatrix} B^{-1}(s|s_0)$$

where

$$B(s) = \begin{bmatrix} \sqrt{\beta(s)} & 0 \\ -\frac{\alpha(s)}{\sqrt{\beta(s)}} & -\frac{1}{\sqrt{\beta(s)}} \end{bmatrix}$$

B is a matrix not magnetic field
↳ SY Lee notation, sorry!

$$B^{-1}(s) = \begin{bmatrix} \frac{1}{\sqrt{\beta(s)}} & 0 \\ \frac{\alpha(s)}{\sqrt{\beta(s)}} & \sqrt{\beta(s)} \end{bmatrix}$$

- Describing ellipse, rotating coordinates
  ↳ more tomorrow

Betatron tune

- Betatron tune is the number of oscillations per turn

$$Q_u = \nu_u = \frac{1}{2\pi} \oint_{s}^{s+C} \frac{ds}{\beta(s)}$$

← circumference

↑ separate for x vs z

- Can have periodic structure within circumference (superperiods) of length L

  i.e. $C = PL$, often define Φ_u as phase advance per period

$$\nu_u = \frac{P\Phi_u}{2\pi} = \frac{1}{2\pi} \oint_{s}^{s+C} \frac{ds}{\beta(s)}$$

- Betatron oscillation frequency is the product $\nu_u f_0$ f_0 is revolution frequency around accelerator